

Hospital Management System - Technical Architecture

Recommended Technology Stack

- Frontend: React.js
- Backend: Node.js with Express
- Database: PostgreSQL
- Caching: Redis
- Cloud Platform: AWS
- Containerization: Docker & Kubernetes

Architecture Overview

The application follows a modular microservices architecture for scalability and maintainability.

Security Architecture

- JWT-based authentication.
- Encrypted patient data storage.
- API gateway protection.
- Role-based authorization.

Scalability Considerations

- Horizontal scaling using Kubernetes.
- Database replication.
- Load balancing for APIs.
- Asynchronous job processing.